

Original practice 1: Charge, measurement and Ohm's law

Original prediction practice—not official questions or a guarantee

Total: 36 marks • Suggested time: 75 minutes

Foundations, meter placement, a series circuit, inverted graph axes and evidence-based Ohm's-law reasoning.

Status and instructions

Original prediction practice—not official questions or a guarantee

Author-created chapter practice, not a claim about an official examination format. All 16 questions are compulsory. Show working and units. Circuit wires, meters and sources are ideal unless stated; resistor temperatures remain constant unless a question explicitly studies a change. Treat stated data as exact unless a tolerance is given.

All case datasets and SVGs are original constructed teaching material, not measurements reported by NCERT. NCERT is cited for concepts and formulas, not as the source of these questions.

- A: Q1-6, $6 \times 1 = 6$ marks
- B: Q7-10, $4 \times 2 = 8$ marks
- C: Q11-13, $3 \times 3 = 9$ marks
- D: Q14, $1 \times 5 = 5$ marks
- E: Q15-16, $2 \times 4 = 8$ marks
- Total: 16 questions, 36 marks

Official curriculum scope and assessment limitation

CBSE Science (086), Class X, 2026-27, PDF p.6 Unit IV lists electric current, potential difference, resistance, resistivity, Ohm's law, resistor combinations, heating and power. Practical 4 on PDF p.7 covers V-I measurements; practical 5 on PDF p.8 covers series and parallel equivalent resistance. Source: https://cbseacademic.nic.in/web_material/CurriculumMain27/SecPart1/Science_SecP1_2026-27.pdf

The same PDF p.6 contains an unresolved teacher-note wording inconsistency naming "Electric Effects of Electric Current" among unassessed topics, despite the explicit Unit IV list. These are chapter study materials, not a guarantee of summative assessment coverage; consult school/CBSE clarification. Unit IV's 13 marks covers a combined unit, not an Electricity-only allotment.

Section A — MCQ

1. [1 marks]

1. A charge counter records 72 C crossing a wire in 3 minutes. What is the mean current?

- A. 24 A
- B. 0.40 A
- C. 216 A
- D. 4.0 A

Source/editorial note: Original prediction practice—not official questions or a guarantee. Original question and worked answer. Concept/formula references: §11.1, printed pp171-172, Eq. (11.1): $I = Q/t$; conventional current opposes electron flow. Official chapter: <https://ncert.nic.in/textbook/pdf/jesc111.pdf>

2. [1 marks]

2. In a metal wire, electrons drift from point M to point N. Which direction describes conventional current in that wire?

- A. M to N
- B. Neither direction
- C. N to M
- D. Both directions simultaneously

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3. [1 marks]

3. A sensor transfers 35 J of electrical energy when 5 C passes through it. What is the potential difference?

- A. 7 V
- B. 175 V
- C. 0.14 V
- D. 30 V

Source/editorial note: Original prediction practice—not official questions or a guarantee. Original question and worked answer. Concept/formula references: §11.2, printed pp173-174, Eqs. (11.2)–(11.3): $V = W/Q$; voltmeter in parallel. Official chapter: <https://ncert.nic.in/textbook/pdf/jesc111.pdf>

4. [1 marks]

4. Which connection measures the potential difference across a test resistor without appreciably changing its current, using ideal meters?

- A. Ammeter across the resistor
- B. Voltmeter in series
- C. Ammeter instead of the resistor
- D. Voltmeter across the resistor

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5. [1 marks]

5. An ohmic resistor carries 0.25 A at 2 V. Its temperature is kept constant. What current flows at 6 V?

- A. 0.083 A
- B. 0.75 A
- C. 1.50 A
- D. 3 A

Source/editorial note: Original prediction practice—not official questions or a guarantee. Original question and worked answer. Concept/formula references: §11.4, printed pp175–177; Activity 11.1 p175; Eqs. (11.4)–(11.7) p176: $V = IR$ at constant temperature; Activity 11.2 pp176–177 compares components. Official chapter: <https://ncert.nic.in/textbook/pdf/jesc111.pdf>

6. [1 marks]

6. An I-versus-V line has current on the vertical axis and voltage on the horizontal axis. Which quantity is its slope for an ohmic resistor?

- A. Resistance R
- B. Power P
- C. Reciprocal resistance $1/R$
- D. Charge Q

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Section B — 2-mark questions

7. [2 marks]

7. A steady 125 mA flows for 80 s. Calculate (a) charge transferred and (b) energy transferred if the potential difference is 4.8 V.

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8. [2 marks]

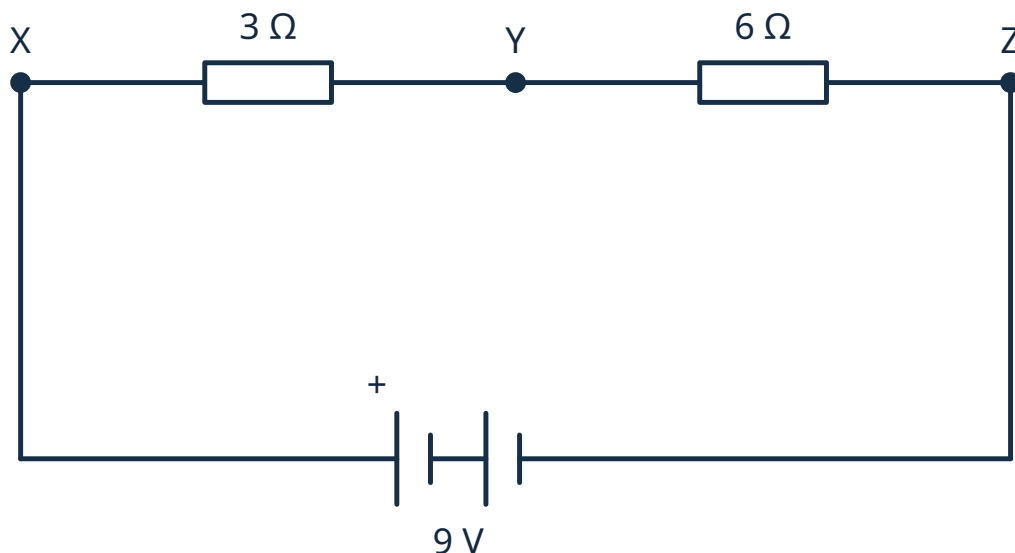
8. Two uniform wires have the same material and temperature. Wire T has three times the length and twice the cross-sectional area of wire S. If S is $8\ \Omega$, calculate T's resistance and state whether their resistivities differ.

Source/editorial note: Original prediction practice—not official questions or a guarantee. Original question and worked answer. Concept/formula references: §11.5, printed pp177–180; Activity 11.3 pp177–178; Eqs. (11.8)–(11.10) p178: $R = \rho l/A$; material and temperature discussion p179. Official chapter: <https://ncert.nic.in/textbook/pdf/jesc111.pdf>

9. [2 marks]

9. In the supplied 9 V circuit, find the current through the $6\ \Omega$ resistor and the potential difference from Y to Z (its magnitude).

9 V series circuit



Source/editorial note: Original prediction practice—not official questions or a guarantee. Original question and worked answer. Concept/formula references: §11.6.1, printed pp182–185; Activity 11.4 p182 (same current); Activity 11.5 p183 (voltage sum); Eqs. (11.11)–(11.14) pp183–184. §11.4, printed pp175–177; Activity 11.1 p175; Eqs. (11.4)–(11.7) p176: $V = IR$ at constant temperature; Activity 11.2 pp176–177 compares components. Official chapter: <https://ncert.nic.in/textbook/pdf/jesc111.pdf>

10. [2 marks]

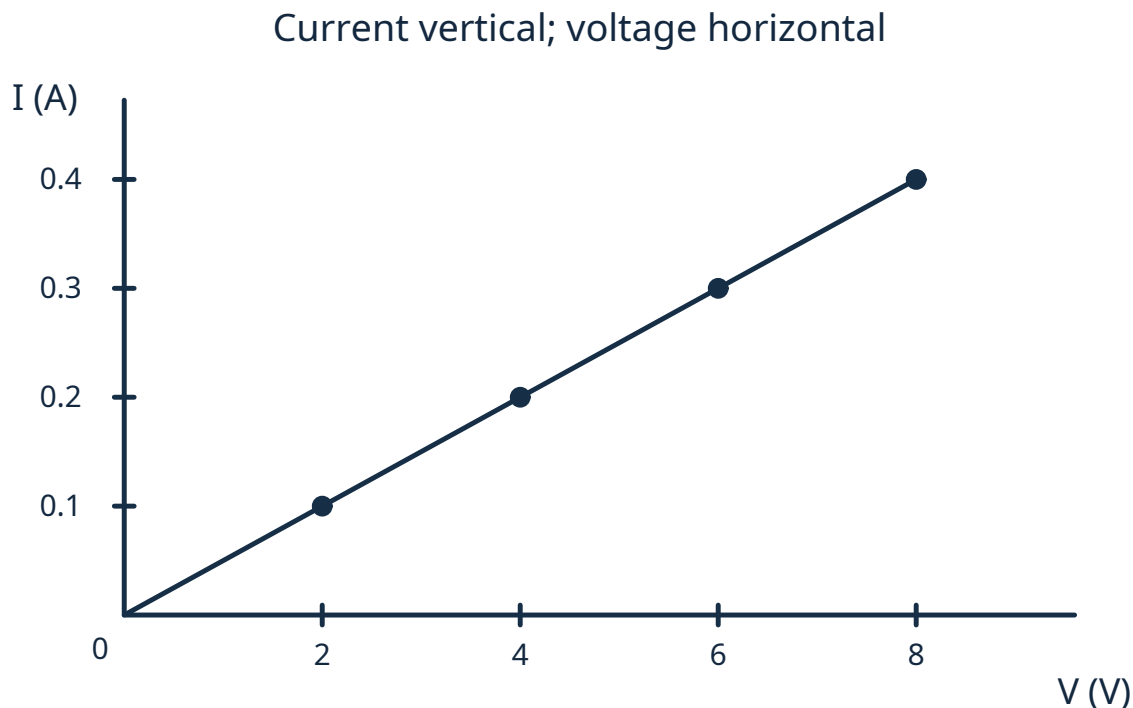
10. A student doubles the voltage across a wire but finds less than twice the current after leaving it connected for a long time. Explain why this alone does not refute Ohm's law, and suggest one measurement improvement.

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Section C — 3-mark questions

11. [3 marks]

11. Use the supplied graph: I is vertical and V is horizontal. (a) Calculate the slope using (2 V, 0.10 A) and (8 V, 0.40 A), with units. (b) Find R . (c) Predict I at 10 V if the same straight-line relation continues.



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12. [3 marks]

12. A uniform laboratory wire has length 1.5 m, cross-sectional area 0.30 mm^2 and resistance 6Ω . (a) Convert its area to m^2 . (b) Calculate resistivity. (c) Find the resistance of a 0.50 m piece of this wire.

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13. [3 marks]

13. A 0.60 A current passes through a resistor for 50 s, transferring 270 J. Determine (a) charge, (b) potential difference and (c) resistance.

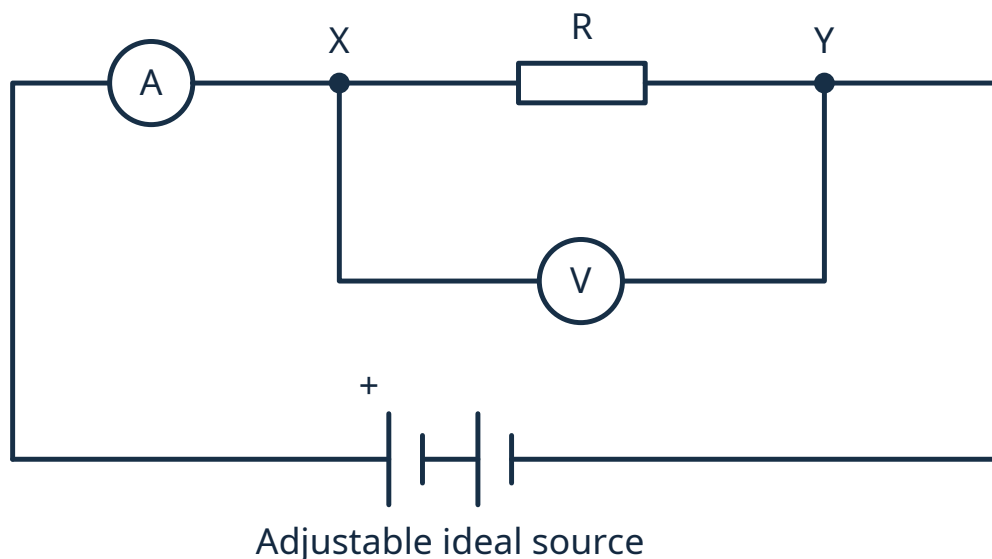
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Section D — 5-mark questions

14. [5 marks]

14. For the supplied measurement circuit: (a) explain why A is in series and V is in parallel (2 marks); (b) a reading gives 4.5 V and 0.30 A: find R (1); (c) predict current at 6 V under controlled temperature (1); (d) name one quantity, other than voltage and current, that must be kept constant when testing Ohm's law (1).

Ohm's-law measurement circuit



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Section E — Case-based questions

15. [4 marks]

15. Constructed case data: a wire is tested at the same temperature. Pairs (V in V, I in A) are (1.2, 0.08), (2.4, 0.16), (3.6, 0.24). (a) Calculate V/I for all three rows (1). (b) State what the ratios support (1). (c) A new entry at 4.8 V says 0.23 A; calculate the current expected from the first three rows (1). (d) Give a justified response to the discrepant entry without claiming the cause is proven (1).

Source/editorial note: Original prediction practice—not official questions or a guarantee. Original question and worked answer. Concept/formula references: §11.4, printed pp175–177; Activity 11.1 p175; Eqs. (11.4)–(11.7) p176: $V = IR$ at constant temperature; Activity 11.2 pp176–177 compares components. Official chapter: <https://ncert.nic.in/textbook/pdf/jesc111.pdf>

16. [4 marks]

16. Constructed case: an ideal 10 V source supplies 4 Ω and 16 Ω in series. A data sheet says current before 4 Ω = 0.50 A, between resistors = 0.35 A, after 16 Ω = 0.50 A; measured drops are 2.0 V and 8.0 V. (a) Identify the inconsistent current entry (1). (b) Calculate the expected current (1). (c) Test the voltage-sum relation (1). (d) Explain why charge is not used up by the first resistor (1).

Source/editorial note: Original prediction practice—not official questions or a guarantee. Original question and worked answer. Concept/formula references: §11.6.1, printed pp182–185; Activity 11.4 p182 (same current); Activity 11.5 p183 (voltage sum); Eqs. (11.11)–(11.14) pp183–184. §11.8, printed pp191–192; Eqs. (11.22)–(11.23) p191: $P = VI = I^2R = V^2/R$; $E = Pt$; 1 kWh = 3.6×10^6 J. Official chapter: <https://ncert.nic.in/textbook/pdf/jesc111.pdf>

Solutions and verification status

Only explicitly verified solutions are printed. A missing solution is not a claim that the question is unsolvable.

Question 1 — original-practice-01-q01

[1 mark] B. $t = 180 \text{ s}$; $I = Q/t = 72/180 = 0.40 \text{ A}$.

Question 2 — original-practice-01-q02

[1 mark] C. Conventional current is opposite to negative-electron motion, so it is from N to M.

Question 3 — original-practice-01-q03

[1 mark] A. $V = W/Q = 35/5 = 7 \text{ V}$.

Question 4 — original-practice-01-q04

[1 mark] D. An ideal voltmeter is connected in parallel across the resistor and draws no current.

Question 5 — original-practice-01-q05

[1 mark] B. $R = 2/0.25 = 8 \Omega$; $I = 6/8 = 0.75 \text{ A}$.

Question 6 — original-practice-01-q06

[1 mark] C. Slope = $\Delta I/\Delta V = 1/R$, not R .

Question 7 — original-practice-01-q07

[1 mark] (a) $I = 0.125 \text{ A}$; $Q = It = 0.125 \times 80 = 10 \text{ C}$.

[1 mark] (b) $W = VQ = 4.8 \times 10 = 48 \text{ J}$.

Question 8 — original-practice-01-q08

[1 mark] $R_T/R_S = (3I/2A)/(I/A) = 3/2$, so $R_T = 12 \Omega$.

[1 mark] Their resistivities are equal: ρ depends on material and temperature, not these dimensions.

Question 9 — original-practice-01-q09

[1 mark] $R_{\text{total}} = 3 + 6 = 9 \Omega$; the common series current is $9/9 = 1 \text{ A}$.

[1 mark] $V_{YZ} = I \times 6 = 6 \text{ V}$.

Question 10 — original-practice-01-q10

[1 mark] The wire may heat and change resistance; Ohm's law requires constant temperature.

[1 mark] Take brief readings, disconnect between readings and allow cooling to the same starting temperature; then compare V/I .

Question 11 — original-practice-01-q11

[1 mark] (a) Slope = $(0.40 - 0.10)/(8 - 2) = 0.050 \text{ A/V}$.

[1 mark] (b) $R = 1/\text{slope} = 20 \Omega$.

[1 mark] (c) $I = V/R = 10/20 = 0.50 \text{ A}$; this is a conditional extrapolation.

Question 12 — original-practice-01-q12

[1 mark] (a) $A = 0.30 \times 10^{-6} = 3.0 \times 10^{-7} \text{ m}^2$.

[1 mark] (b) $\rho = RA/l = 6 \times 3.0 \times 10^{-7}/1.5 = 1.2 \times 10^{-6} \Omega \text{ m}$.

[1 mark] (c) Same area and material: $R_{\text{piece}} = 6 \times (0.50/1.5) = 2 \Omega$.

Question 13 — original-practice-01-q13

[1 mark] (a) $Q = It = 0.60 \times 50 = 30 \text{ C}$.

[1 mark] (b) $V = W/Q = 270/30 = 9 \text{ V}$.

[1 mark] (c) $R = V/I = 9/0.60 = 15 \Omega$.

Question 14 — original-practice-01-q14

[1 mark] (a) A is in series so the entire resistor current passes through it.

[1 mark] (a) V is across the resistor's two endpoints to measure their potential difference; ideally it draws no current.

[1 mark] (b) $R = 4.5/0.30 = 15 \Omega$.

[1 mark] (c) $I = 6/15 = 0.40 \text{ A}$.

[1 mark] (d) Keep the wire temperature constant (also retain the same wire geometry and material).

Question 15 — original-practice-01-q15

[1 mark] (a) $1.2/0.08 = 2.4/0.16 = 3.6/0.24 = 15 \Omega$.

[1 mark] (b) Constant V/I supports ohmic behaviour over the tested range at the controlled temperature.

[1 mark] (c) $I_{\text{expected}} = 4.8/15 = 0.32 \text{ A}$.

[1 mark] (d) Recheck meter reading/range, connections and temperature, then repeat; the discrepancy alone does not establish its cause.

Question 16 — original-practice-01-q16

[1 mark] (a) The middle 0.35 A entry is inconsistent: an unbranched steady circuit has equal current at all positions.

[1 mark] (b) $R = 4 + 16 = 20 \Omega$, so $I = 10/20 = 0.50 \text{ A}$ everywhere.

[1 mark] (c) $2.0 + 8.0 = 10.0 \text{ V}$, equal to the source voltage, as required.

[1 mark] (d) Charges circulate; electrical energy is transferred to the resistor, but charge is not consumed.